THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 2: **Invisible Living World: Beyond Our Naked Eye** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A mouse and an elephant have cells of similar size, yet the elephant is far larger. This shows that: 1

(a) cell size depends on the age of the animal
(b) elephant cells are about a thousand times bigger
(c) bigger organisms have more cells, not bigger cells
(d) a mouse has more cells than an elephant

2. Assertion (A): Nerve cells are long and branched. Reason (R): This shape helps in carrying messages quickly. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

3. Which among these cannot be seen under a school microscope as a distinct cell? 1

(a) virus **(b)** cheek cell **(c)** Paramecium **(d)** onion cell

4. A mushroom is classified with fungi and not plants because it: 1

(a) has leaves **(b)** has roots
(c) has green colour **(d)** lacks chlorophyll and cannot make food

5. Assertion (A): A multicellular organism has organs. Reason (R): A unicellular organism has organs for each function. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

6. A tissue is damaged. This can reduce the function of the: 1

(a) no organ **(b)** an unrelated system
(c) organ it belongs to **(d)** only the cell wall

7. Pathogenic bacteria differ from useful bacteria in that they: 1

(a) are bigger **(b)** have no cell membrane
(c) cause disease in host organisms **(d)** have chloroplasts

8. Assertion (A): Yeast is called a fungus. Reason (R): It carries out photosynthesis with chlorophyll. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

9. **Assertion (A):** Bacteria are prokaryotic organisms. 1
Reason (R): Bacteria are single-celled organisms.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** The nucleus of a cell controls its activities. 1
Reason (R): The nucleus is present in every living cell without exception.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. State two differences between an organelle and an organ. 2

12. Why does a cell need a membrane even when it has a cell wall? 2

13. State the importance of stain and cover slip in preparing a good temporary mount. 2

14. Why do cells of leaves outnumber root cells in the number of chloroplasts? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Estimate: the width of a human hair is 0.07 mm. A cell is 20 micrometres. How many cells fit across the hair? 3

16. A student's drawing of a plant cell has no vacuole and no wall. List the errors and corrections. 3

17. List three safety rules for handling a microscope and slides. 3

OR

A bacterial colony doubles every 20 minutes. Starting from one bacterium, how many in 2 hours? 3

18. Explain how bacteria can be seen under a school microscope and what stain helps. 3

OR

How did the discovery of microbes help in preventing the spread of disease? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A microscope has an eyepiece lens of magnification 10x and an objective lens of 40x. Arjun views a red blood cell of diameter about 7 micrometres, a bacterium of about 2 micrometres, and tries to see a virus of about 0.1 micrometre. He can see the first two but the virus cannot be seen clearly. 5
(i) Find the total magnification of the microscope.
(ii) Which is the largest of the three?
(iii) Name a part of a plant cell that stores water and dissolved substances.
(iv) Why can the virus not be seen clearly with this microscope?

20. Compare the structure of bacteria, fungi and protozoa with an example of each and one way they affect humans. 5

OR

Discuss how scientists studied the invisible world with the help of the microscope across time: simple lens to compound microscope, and a limit of the light microscope. 5

END OF QUESTION PAPER — Total: 40 Marks